

2) Respuesta de módulo y fase. ($z = e^{j\Omega}$)

a $T(z) \Big|_{z=e^{j\Omega}} = T(\Omega) = e^{-j3\Omega} + e^{-j2\Omega} + e^{-j\Omega} + 1 \cdot e^0$

$$= e^{-j3/2\Omega} \cdot (e^{-j3/2\Omega} + e^{-j1/2\Omega} + e^{j1/2\Omega} + e^{j3/2\Omega})$$

$$= e^{-j3/2\Omega} \cdot [2 \cos(3/2\Omega) + 2 \cos(1/2\Omega)]$$

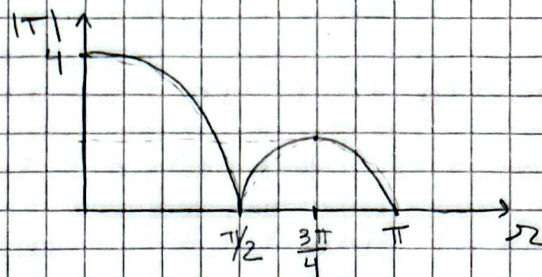
$$T(\Omega) = \underbrace{[2 \cos(3/2\Omega) + 2 \cos(1/2\Omega)]}_{= T_R(\Omega)} \cdot \underbrace{e^{-j3/2\Omega}}_{\neq T(\Omega)}$$

$$|T(\Omega)| = |T_R(\Omega)|$$

• Respuesta de módulo

$$|T(\omega)| = 2 \cdot |\cos(3/2 \omega) + \cos(1/2 \omega)|$$

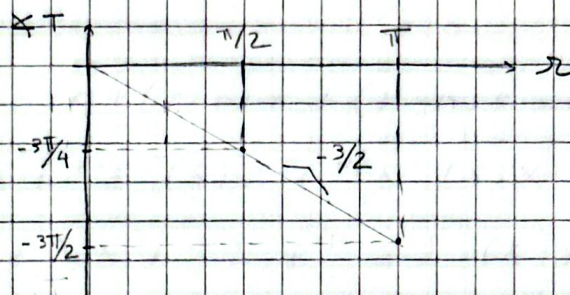
$$\begin{aligned} \omega = 0: |T| &= 4 \\ \omega = \pi/2: |T| &= 0 \\ \omega = 3\pi/4: |T| &\approx 1,08 \\ \omega = \pi: |T| &= 0 \end{aligned}$$



• Respuesta de fase

$$\angle T(\omega) = -3/2 \omega$$

$$\begin{aligned} \omega = 0: \angle T &= 0 \\ \omega = \pi/2: \angle T &= -3\pi/4 \\ \omega = \pi: \angle T &= -3\pi/2 \end{aligned}$$



b)
$$\begin{aligned} T(\omega) &= e^{-j4\omega} + e^{-j3\omega} + e^{-j2\omega} + e^{-j\omega} + e^0 \\ &= e^{-j2\omega} \cdot (e^{-j2\omega} + e^{-j\omega} + e^0 + e^{j\omega} + e^{j2\omega}) \\ &= e^{-j2\omega} \cdot [2 \cdot \cos(2\omega) + 2 \cdot \cos(\omega) + 1] \\ T(\omega) &= \underbrace{[2 \cdot (\cos(2\omega) + \cos(\omega)) + 1]}_{= T_R(\omega)} \cdot e^{j2\omega} \end{aligned}$$

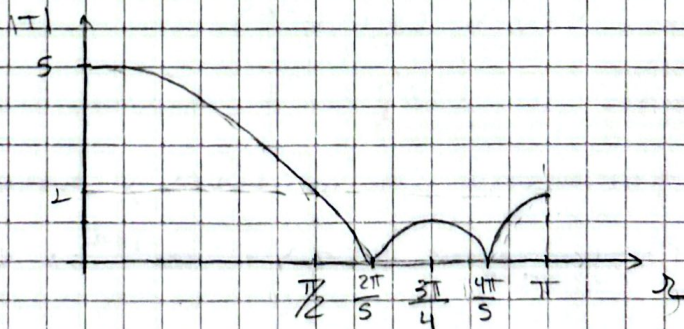
$\angle T(\omega)$

$\hookrightarrow |T(\omega)| = |T_R(\omega)|$

• Respuesta de módulo

$$|T(\omega)| = |2 \cdot (\cos(2\omega) + \cos(\omega)) + 1|$$

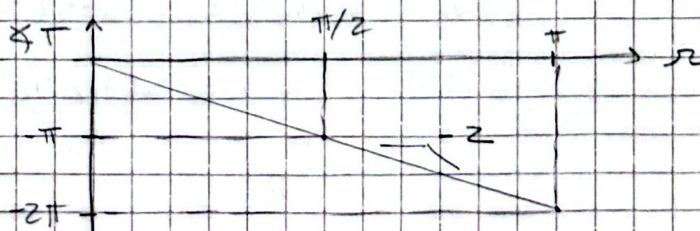
$$\begin{aligned} \omega = 0: |T| &= 5 \\ \omega = \pi/2: |T| &= 1 \\ \omega = 3\pi/4: |T| &\approx 0,414 \\ \omega = 2\pi/5: |T| &= 0 \\ \omega = 4\pi/5: |T| &= 0 \\ \omega = \pi: |T| &= 1 \end{aligned}$$



• Respuesta de fase

$$\angle T(\omega) = -2\omega$$

$$\begin{aligned} \omega = 0: \angle T &= 0 \\ \omega = \pi/2: \angle T &= -\pi \\ \omega = \pi: \angle T &= -2\pi \end{aligned}$$



$$c. T(\omega) = e^0 - e^{-j\omega} = e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})$$

$$= e^{-j\omega/2} (j \cdot 2 \sin(\omega/2))$$

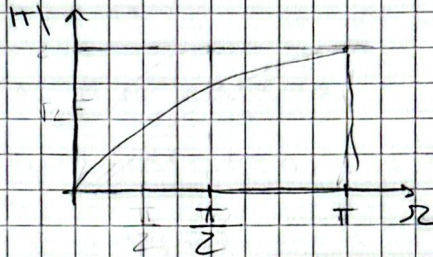
$$T(\omega) = \underbrace{[j \cdot 2 \sin(\omega/2)]}_{= T_R(\omega)} \cdot e^{j\omega/2} \rightarrow \angle T(\omega)$$

$$|T(\omega)| = |T_R(\omega)|$$

• Respuesta de módulo

$$|T(\omega)| = 2 \cdot |\sin(\omega/2)|$$

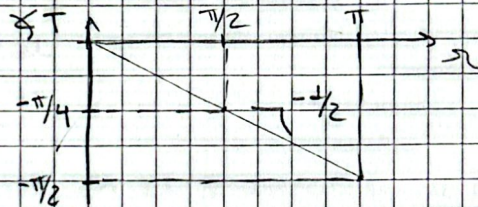
$$\begin{aligned} \omega = 0 &: |T| = 0 \\ \omega = \pi/2 &: |T| = \sqrt{2} \\ \omega = \pi &: |T| = 2 \\ \omega = 3\pi/4 &: |T| \approx 1.85 \\ \omega = 5\pi/8 &: |T| \approx 1.66 \end{aligned}$$



• Respuesta de fase

$$\angle T(\omega) = -\omega/2$$

$$\begin{aligned} \omega = 0 &: \angle T = 0 \\ \omega = \pi/2 &: \angle T = -\pi/4 \\ \omega = \pi &: \angle T = -\pi/2 \end{aligned}$$



$$d. T(\omega) = e^0 - e^{-j2\omega} = e^{-j\omega} (e^{j\omega} - e^{-j\omega})$$

$$= e^{-j\omega} \cdot j \cdot 2 \sin(\omega)$$

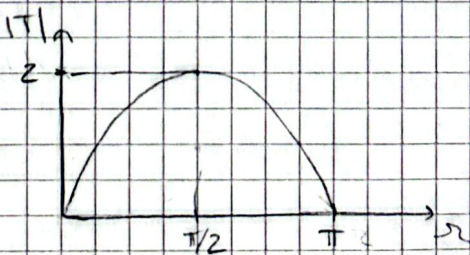
$$T(\omega) = \underbrace{(j \cdot 2 \sin(\omega))}_{= T_R(\omega)} \cdot e^{-j\omega} \rightarrow \angle T(\omega)$$

$$|T(\omega)| = |T_R(\omega)|$$

• Respuesta de módulo

$$|T(\omega)| = 2 \cdot |\sin(\omega)|$$

$$\begin{aligned} \omega = 0 &: |T| = 0 \\ \omega = \pi/2 &: |T| = 2 \\ \omega = \pi &: |T| = 0 \end{aligned}$$



• Respuesta de fase

$$\angle T(\omega) = -\omega$$

